

UNIT II

Combinational Logic: MSI devices like Magnitude comparator, Multiplexers, Demultiplexers, Decoders, Encoders. Multiplexed display, half and full adders, subtractors, serial and parallel adders, BCD adder

① What is a Combinational Circuit?

A combinational circuit is a digital circuit in which:

- Output depends only on **current inputs**
- It has **no memory**
- No clock is needed

If input changes → output changes immediately.

Example:

Calculator addition

If you change $5 + 3$ to $6 + 3$, result changes instantly.

Difference from Sequential Circuit

Combinational	Sequential
No memory	Has memory
No clock	Needs clock
Output = present input	Output = present + past

② MSI Devices (Medium Scale Integration)

MSI devices are ICs that contain multiple logic gates inside.

Examples in syllabus:

- Magnitude Comparator
- Multiplexer
- Demultiplexer
- Decoder
- Encoder

These are building blocks of CPU and digital systems.

3 Magnitude Comparator

Purpose

It compares two binary numbers and tells:

- $A > B$
- $A = B$
- $A < B$

1-Bit Comparator

Inputs:

A, B

Outputs:

- $A > B$
- $A = B$
- $A < B$

Truth Table

A	B	$A > B$	$A = B$	$A < B$
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

Logic Equations

$$A > B = A B'$$

$$A < B = A' B$$

$$A = B = A' B' + A B$$

Multi-bit Comparator

If comparing 2-bit numbers:

$$A = A_1 A_0$$

$$B = B_1 B_0$$

First compare A_1 and B_1 (MSB).

If equal $\rightarrow$ compare A_0 and B_0 .

Where Used?

- ALU compare instruction
- Sorting circuits
- Digital decision making

4 Multiplexer (MUX)

Definition

A multiplexer selects one input out of many inputs and sends it to output.

Also called:

Data Selector

4:1 MUX

Inputs:

I0, I1, I2, I3

Select lines:

S1, S0

Output:

Y

Truth Table

S1	S0	Output
0	0	I0
0	1	I1
1	0	I2
1	1	I3

Expression

$$Y = I_0 S_1' S_0' + I_1 S_1' S_0 + I_2 S_1 S_0' + I_3 S_1 S_0$$

Example

Suppose:

I0=0, I1=1, I2=0, I3=1

S1S0 = 01

Output = I1 = 1

Where Used?

- CPU data bus selection

- Selecting register output
- Implementing Boolean functions

Think of it like a railway switch — only one track is selected.

5 Demultiplexer (DEMUX)

Opposite of MUX.

One input → many outputs.

1:4 DEMUX

Input:

D

Select lines:

S1, S0

Outputs:

Y0, Y1, Y2, Y3

Only one output gets input D.

Example

If:

D=1

S1S0 = 10

Then Y2 = 1

Other outputs = 0

Used In

- Data distribution
- Memory systems
- Communication

6 Decoder

Definition

A decoder converts binary input into one active output line.

If input is n bits → output is 2^n lines.

2:4 Decoder

Inputs:

A, B

Outputs:

D0, D1, D2, D3

Truth Table

A	B	Active Output
0	0	D0
0	1	D1
1	0	D2
1	1	D3

Only one output is HIGH at a time.

Used In

- Memory address decoding
- Instruction decoding
- 7-segment display

CPU uses decoder to understand instructions.

7 Encoder

Opposite of decoder.

Many inputs → few outputs.

4:2 Encoder

If D2 = 1 → Output = 10

Truth Table

D3 D2 D1 D0	Output
0001	00
0010	01
0100	10
1000	11

Priority Encoder

If multiple inputs active → highest priority selected.

Used in:

- Interrupt systems

8 Multiplexed Display

Used to control multiple 7-segment displays using fewer pins.

Working

- Turn ON one display at a time
- Switch very fast
- Human eye sees continuous number

Used in:

- Digital clock
- Calculator
- Embedded board

9 Half Adder

Adds two bits.

Inputs:

A, B

Outputs:

Sum, Carry

Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Formula

$$\text{Sum} = A \oplus B$$

$$\text{Carry} = AB$$

10 Full Adder

Adds three bits:

A, B, Cin

Outputs:

Sum, Cout

Formulas

$$\text{Sum} = A \oplus B \oplus \text{Cin}$$

$$\text{Cout} = AB + B\text{Cin} + A\text{Cin}$$

Why Important?

Multi-bit addition is done by connecting many full adders.

This forms:

Parallel Adder (used in CPU)

1 1 Half Subtractor

Subtracts two bits.

Inputs:

A, B

$$\text{Difference} = A \oplus B$$

$$\text{Borrow} = A'B$$

1 2 Full Subtractor

Inputs:

A, B, Bin

$$\text{Difference} = A \oplus B \oplus \text{Bin}$$

$$\text{Borrow} = A'B + A'\text{Bin} + B\text{Bin}$$

1 3 Serial Adder

Uses:

- One full adder
- Shift registers

Adds bit-by-bit.

Advantage:

Less hardware

Disadvantage:

Slow

1 4 Parallel Adder

Uses multiple full adders.

Adds all bits at same time.

Advantage:

Fast

Used in:

- Processors
- High-speed systems

1 5 BCD Adder

Adds two BCD numbers.

Step 1:

Add normally using 4-bit binary adder.

Step 2:

If result > 1001 (9), add 0110 (6).

Why add 6?

Because 1010–1111 are invalid BCD codes.

Example

9 + 8

1001

+1000

10001

Result invalid → Add 6

Final BCD result = 17

1 Combinational vs Sequential Circuit

Feature	Combinational Circuit	Sequential Circuit
Memory	✗ No memory	✓ Has memory
Clock Required	✗ No	✓ Yes
Output Depends On	Present input only	Present + past input
Feedback	No feedback	Feedback present

Example	Adder, MUX	Flip-flop, Counter
Speed	Fast (instant output)	Depends on clock

★ Exam tip: Always write “no memory” and “no clock” for combinational.

2 Multiplexer (MUX) vs Demultiplexer (DEMUX)

Feature	MUX	DEMUX
Full Name	Multiplexer	Demultiplexer
Function	Many inputs → One output	One input → Many outputs
Also Called	Data Selector	Data Distributor
Control	Select lines choose input	Select lines choose output
Used In	CPU bus selection	Memory systems
Example	4:1 MUX	1:4 DEMUX

Simple idea:

MUX selects one line.

DEMUX distributes one line.

3 Decoder vs Encoder

Feature	Decoder	Encoder
Function	Binary input → Many outputs	Many inputs → Binary output
Input Lines	n	2^n
Output Lines	2^n	n
Active Output	Only one active	One output code generated
Used In	Memory selection	Interrupt system
Example	2:4 Decoder	4:2 Encoder

Remember:

Decoder increases lines.

Encoder reduces lines.

4 Encoder vs Priority Encoder

Feature	Normal Encoder	Priority Encoder
Multiple Active Inputs	Not allowed	Allowed
Output	Undefined if multiple active	Highest priority selected
Used In	Simple circuits	Interrupt controller
Reliability	Less	More practical

5 Half Adder vs Full Adder

Feature	Half Adder	Full Adder
Inputs	2 (A,B)	3 (A,B,Cin)
Carry Input	✗ No	✓ Yes
Outputs	Sum, Carry	Sum, Carry
Equation (Sum)	$A \oplus B$	$A \oplus B \oplus Cin$
Used For	Single-bit addition	Multi-bit addition
Used In CPU	No	Yes

★ Very important difference.

6 Half Subtractor vs Full Subtractor

Feature	Half Subtractor	Full Subtractor
Inputs	2 (A,B)	3 (A,B,Bin)
Borrow Input	✗ No	✓ Yes
Outputs	Difference, Borrow	Difference, Borrow
Used For	Single subtraction	Multi-bit subtraction

7 Serial Adder vs Parallel Adder

Feature	Serial Adder	Parallel Adder
Hardware Used	One full adder	Multiple full adders
Speed	Slow	Fast
Cost	Low	High
Addition Type	Bit by bit	All bits at once
Used In	Low-cost systems	Processors

Exam line:

Serial = less hardware, more time.

Parallel = more hardware, less time.

8 Binary Adder vs BCD Adder

Feature	Binary Adder	BCD Adder
Adds	Binary numbers	BCD digits
Correction Step	Not required	Add 6 if sum > 9
Used In	CPU	Calculators
Complexity	Simple	More complex

Important:

In BCD adder → correction step is compulsory.

9 Comparator vs Decoder

Feature	Comparator	Decoder
Purpose	Compare numbers	Select output line
Outputs	$A > B$, $A = B$, $A < B$	One active output
Used In	ALU compare	Memory selection

10 MUX vs Decoder

Feature	MUX	Decoder
Inputs	Data inputs + select	Only binary input
Output	One output	Multiple outputs
Purpose	Select data	Generate output line
Control	Select lines	Input lines